General tissue texture: Spatial split (held-out region of a piece seen in training; within-specimen interpolation) (1/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	+ A stained refs (extra input)	+ B unstained context	+ specimen label (oracle)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix
A row3 col5 10x10 brain			LPIPS 0.410 PSNR 12.3 SSIM 0.24	LPIPS 0.342 PSNR 12.8 SSIM 0.31	LPIPS 0.481 PSNR 11.2 SSIM 0.29	LPIPS 0.346 PSNR 13.0 SSIM 0.34	LPIPS 0.586 PSNR 11.4 SSIM 0.11	LPIPS 0.671 PSNR 9.2 SSIM 0.34	LPIPS 0.318 PSNR 13.3 SSIM 0.36	LPIPS 1.032 PSNR 6.5 SSIM 0.33	LPIPS 0.868 PSNR 10.1 SSIM 0.36
D row6 col8 10x10 brain			LPIPS 0.579 PSNR 19.8 SSIM 0.45	LPIPS 0.560 PSNR 20.6 SSIM 0.46	LPIPS 0.560 PSNR 19.0 SSIM 0.44	LPIPS 0.530 PSNR 23.6 SSIM 0.52	LPIPS 0.621 PSNR 17.5 SSIM 0.30	LPIPS 0.804 PSNR 14.7 SSIM 0.38	LPIPS 0.783 PSNR 13.5 SSIM 0.21	LPIPS 0.749 PSNR 20.8 SSIM 0.55	LPIPS 0.567 PSNR 22.9 SSIM 0.52
E row2 col4 10x10 brain			LPIPS 0.529 PSNR 18.6 SSIM 0.42	LPIPS 0.502 PSNR 16.8 SSIM 0.39	LPIPS 0.719 PSNR 17.8 SSIM 0.43	LPIPS 0.495 PSNR 18.0 SSIM 0.42	LPIPS 0.567 PSNR 18.0 SSIM 0.29	LPIPS 0.598 PSNR 15.7 SSIM 0.39	LPIPS 0.615 PSNR 16.2 SSIM 0.44	LPIPS 0.965 PSNR 10.8 SSIM 0.34	LPIPS 0.717 PSNR 18.6 SSIM 0.43
G row7 col1 10x10 brain			LPIPS 0.460 PSNR 16.3 SSIM 0.46	LPIPS 0.471 PSNR 16.8 SSIM 0.45	LPIPS 0.486 PSNR 16.7 SSIM 0.46	LPIPS 0.479 PSNR 15.9 SSIM 0.48	LPIPS 0.504 PSNR 16.4 SSIM 0.45	LPIPS 0.681 PSNR 13.8 SSIM 0.38	LPIPS 0.610 PSNR 15.2 SSIM 0.46	LPIPS 1.046 PSNR 8.8 SSIM 0.37	LPIPS 0.501 PSNR 15.8 SSIM 0.52
G row8 col6 10x10 brain			LPIPS 0.473 PSNR 16.3 SSIM 0.36	LPIPS 0.455 PSNR 15.8 SSIM 0.32	LPIPS 0.457 PSNR 15.7 SSIM 0.33	LPIPS 0.434 PSNR 16.5 SSIM 0.33	LPIPS 0.616 PSNR 12.4 SSIM 0.14	LPIPS 0.707 PSNR 14.6 SSIM 0.20	LPIPS 0.694 PSNR 12.4 SSIM 0.36	LPIPS 0.777 PSNR 8.4 SSIM 0.34	LPIPS 0.638 PSNR 18.1 SSIM 0.40

Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

General tissue texture: Spatial split (held-out region of a piece seen in training; within-specimen interpolation) (2/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	+ A stained refs (extra input)	+ B unstained context	+ specimen label (oracle)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix
H row2 col8 10x10 brain			LPIPS 0.604 PSNR 17.1 SSIM 0.44	LPIPS 0.529 PSNR 19.1 SSIM 0.45	LPIPS 0.550 PSNR 19.6 SSIM 0.46	LPIPS 0.579 PSNR 20.6 SSIM 0.49	LPIPS 0.662 PSNR 19.0 SSIM 0.41	LPIPS 0.553 PSNR 19.6 SSIM 0.43	LPIPS 0.561 PSNR 18.6 SSIM 0.52	LPIPS 1.055 PSNR 10.9 SSIM 0.34	LPIPS 0.410 PSNR 21.0 SSIM 0.49
I row7 col0 10x10 brain			LPIPS 0.352 PSNR 20.3 SSIM 0.70	LPIPS 0.403 PSNR 15.6 SSIM 0.68	LPIPS 0.357 PSNR 19.1 SSIM 0.52	LPIPS 0.374 PSNR 18.2 SSIM 0.69	LPIPS 0.395 PSNR 17.7 SSIM 0.49	LPIPS 0.679 PSNR 12.7 SSIM 0.44	LPIPS 0.736 PSNR 14.3 SSIM 0.41	LPIPS 0.888 PSNR 11.0 SSIM 0.43	LPIPS 0.531 PSNR 18.7 SSIM 0.51
B row6 col3 10x10 heart			LPIPS 0.492 PSNR 15.2 SSIM 0.29	LPIPS 0.477 PSNR 15.2 SSIM 0.27	LPIPS 0.440 PSNR 13.8 SSIM 0.23	LPIPS 0.412 PSNR 15.5 SSIM 0.35	LPIPS 0.520 PSNR 15.5 SSIM 0.26	LPIPS 0.664 PSNR 13.5 SSIM 0.32	LPIPS 0.469 PSNR 15.2 SSIM 0.22	LPIPS 0.774 PSNR 18.8 SSIM 0.41	LPIPS 0.626 PSNR 16.5 SSIM 0.38
C row3 col4 10x10 heart			LPIPS 0.458 PSNR 14.9 SSIM 0.29	LPIPS 0.514 PSNR 17.4 SSIM 0.34	LPIPS 0.600 PSNR 16.0 SSIM 0.36	LPIPS 0.464 PSNR 17.1 SSIM 0.38	LPIPS 0.707 PSNR 15.8 SSIM 0.33	LPIPS 0.678 PSNR 14.7 SSIM 0.38	LPIPS 0.670 PSNR 17.5 SSIM 0.40	LPIPS 0.874 PSNR 10.6 SSIM 0.33	LPIPS 0.546 PSNR 20.6 SSIM 0.43
F row3 col7 10x10 heart			LPIPS 0.586 PSNR 19.5 SSIM 0.44	LPIPS 0.601 PSNR 17.3 SSIM 0.41	LPIPS 0.507 PSNR 19.9 SSIM 0.49	LPIPS 0.479 PSNR 21.0 SSIM 0.53	LPIPS 0.539 PSNR 20.5 SSIM 0.49	LPIPS 0.803 PSNR 13.7 SSIM 0.36	LPIPS 0.608 PSNR 17.2 SSIM 0.42	LPIPS 0.880 PSNR 10.5 SSIM 0.34	LPIPS 0.453 PSNR 20.6 SSIM 0.60

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General tissue texture: Spatial split (held-out region of a piece seen in training; within-specimen interpolation) (3/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	+ A stained refs (extra input)	+ B unstained context	+ specimen label (oracle)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix
J row0 col4 5x5 heart			LPIPS 0.597 PSNR 19.4 SSIM 0.47	LPIPS 0.431 PSNR 19.6 SSIM 0.46	LPIPS 0.442 PSNR 19.5 SSIM 0.52	LPIPS 0.529 PSNR 18.8 SSIM 0.50	LPIPS 0.563 PSNR 18.4 SSIM 0.43	LPIPS 0.876 PSNR 12.4 SSIM 0.38	LPIPS 0.762 PSNR 17.4 SSIM 0.51	LPIPS 0.946 PSNR 9.6 SSIM 0.34	LPIPS 0.744 PSNR 13.5 SSIM 0.48
J row1 col3 5x5 heart			LPIPS 0.711 PSNR 17.2 SSIM 0.33	LPIPS 0.625 PSNR 17.8 SSIM 0.47	LPIPS 0.522 PSNR 19.6 SSIM 0.42	LPIPS 0.537 PSNR 20.9 SSIM 0.45	LPIPS 0.528 PSNR 19.7 SSIM 0.35	LPIPS 0.859 PSNR 14.2 SSIM 0.41	LPIPS 0.886 PSNR 14.5 SSIM 0.39	LPIPS 1.016 PSNR 11.0 SSIM 0.34	LPIPS 0.655 PSNR 17.9 SSIM 0.45
J row5 col10 10x10 heart			LPIPS 0.618 PSNR 15.9 SSIM 0.33	LPIPS 0.469 PSNR 16.8 SSIM 0.46	LPIPS 0.469 PSNR 17.8 SSIM 0.43	LPIPS 0.480 PSNR 16.2 SSIM 0.48	LPIPS 0.500 PSNR 18.9 SSIM 0.34	LPIPS 0.683 PSNR 13.6 SSIM 0.40	LPIPS 0.770 PSNR 15.4 SSIM 0.42	LPIPS 1.129 PSNR 8.0 SSIM 0.33	LPIPS 0.643 PSNR 15.6 SSIM 0.45
K row1 col1 10x10 heart			LPIPS 0.558 PSNR 18.1 SSIM 0.51	LPIPS 0.450 PSNR 19.1 SSIM 0.60	LPIPS 0.435 PSNR 18.0 SSIM 0.47	LPIPS 0.484 PSNR 17.5 SSIM 0.60	LPIPS 0.503 PSNR 18.0 SSIM 0.40	LPIPS 0.769 PSNR 13.7 SSIM 0.44	LPIPS 0.555 PSNR 13.8 SSIM 0.41	LPIPS 0.891 PSNR 10.3 SSIM 0.34	LPIPS 0.624 PSNR 17.0 SSIM 0.50
K row3 col6 10x10 heart			LPIPS 0.528 PSNR 16.9 SSIM 0.50	LPIPS 0.445 PSNR 18.0 SSIM 0.55	LPIPS 0.400 PSNR 18.2 SSIM 0.52	LPIPS 0.387 PSNR 17.7 SSIM 0.61	LPIPS 0.505 PSNR 18.4 SSIM 0.42	LPIPS 0.740 PSNR 14.6 SSIM 0.45	LPIPS 0.743 PSNR 12.9 SSIM 0.53	LPIPS 0.876 PSNR 11.4 SSIM 0.34	LPIPS 0.552 PSNR 20.0 SSIM 0.57

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Biological features: Spatial split (held-out region of a piece seen in training; within-specimen interpolation) (1/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	+ A stained refs (extra input)	+ B unstained context	+ specimen label (oracle)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix
A row7 col5 10x10 brain			LPIPS 0.771 PSNR 9.3 SSIM 0.35	LPIPS 0.535 PSNR 11.0 SSIM 0.29	LPIPS 0.529 PSNR 11.3 SSIM 0.27	LPIPS 0.511 PSNR 11.0 SSIM 0.34	LPIPS 0.836 PSNR 9.3 SSIM 0.27	LPIPS 0.609 PSNR 9.2 SSIM 0.30	LPIPS 0.619 PSNR 9.1 SSIM 0.29	LPIPS 0.868 PSNR 10.6 SSIM 0.37	LPIPS 0.712 PSNR 9.7 SSIM 0.37
D (Zpart1) row1 col2 5x5 brain			LPIPS 0.601 PSNR 16.5 SSIM 0.50	LPIPS 0.621 PSNR 17.4 SSIM 0.49	LPIPS 0.687 PSNR 15.1 SSIM 0.44	LPIPS 0.508 PSNR 21.1 SSIM 0.57	LPIPS 0.614 PSNR 17.6 SSIM 0.43	LPIPS 0.785 PSNR 14.0 SSIM 0.42	LPIPS 0.697 PSNR 20.5 SSIM 0.51	LPIPS 0.633 PSNR 22.5 SSIM 0.68	LPIPS 0.568 PSNR 22.2 SSIM 0.60
E row6 col10 10x10 brain			LPIPS 0.572 PSNR 18.2 SSIM 0.55	LPIPS 0.530 PSNR 17.2 SSIM 0.51	LPIPS 0.530 PSNR 17.8 SSIM 0.53	LPIPS 0.508 PSNR 17.9 SSIM 0.52	LPIPS 0.649 PSNR 17.0 SSIM 0.46	LPIPS 0.592 PSNR 18.6 SSIM 0.34	LPIPS 0.576 PSNR 18.4 SSIM 0.54	LPIPS 0.905 PSNR 8.5 SSIM 0.34	LPIPS 0.569 PSNR 20.7 SSIM 0.56
G row1 col1 10x10 brain			LPIPS 0.546 PSNR 13.5 SSIM 0.36	LPIPS 0.528 PSNR 14.6 SSIM 0.35	LPIPS 0.554 PSNR 13.6 SSIM 0.38	LPIPS 0.497 PSNR 15.2 SSIM 0.37	LPIPS 0.636 PSNR 10.8 SSIM 0.25	LPIPS 0.724 PSNR 10.4 SSIM 0.35	LPIPS 0.653 PSNR 13.6 SSIM 0.37	LPIPS 0.954 PSNR 7.7 SSIM 0.34	LPIPS 0.688 PSNR 11.1 SSIM 0.40
G row3 col2 5x5 brain			LPIPS 0.544 PSNR 14.4 SSIM 0.32	LPIPS 0.611 PSNR 14.8 SSIM 0.36	LPIPS 0.594 PSNR 15.4 SSIM 0.34	LPIPS 0.602 PSNR 13.9 SSIM 0.39	LPIPS 0.567 PSNR 15.2 SSIM 0.26	LPIPS 0.795 PSNR 12.8 SSIM 0.35	LPIPS 0.802 PSNR 13.4 SSIM 0.40	LPIPS 1.137 PSNR 8.2 SSIM 0.34	LPIPS 0.633 PSNR 13.9 SSIM 0.40

Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

Biological features: Spatial split (held-out region of a piece seen in training; within-specimen interpolation) (2/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	+ A stained refs (extra input)	+ B unstained context	+ specimen label (oracle)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix
G row4 col5 10x10 brain			LPIPS 0.578 PSNR 14.9 SSIM 0.34	LPIPS 0.685 PSNR 13.6 SSIM 0.32	LPIPS 0.665 PSNR 11.7 SSIM 0.31	LPIPS 0.576 PSNR 13.1 SSIM 0.36	LPIPS 0.702 PSNR 15.4 SSIM 0.27	LPIPS 0.676 PSNR 14.1 SSIM 0.34	LPIPS 0.573 PSNR 10.4 SSIM 0.29	LPIPS 0.723 PSNR 18.5 SSIM 0.43	LPIPS 0.574 PSNR 17.8 SSIM 0.42
G row5 col6 10x10 brain			LPIPS 0.456 PSNR 17.1 SSIM 0.35	LPIPS 0.484 PSNR 16.7 SSIM 0.33	LPIPS 0.484 PSNR 14.9 SSIM 0.32	LPIPS 0.518 PSNR 13.3 SSIM 0.37	LPIPS 0.531 PSNR 17.4 SSIM 0.27	LPIPS 0.750 PSNR 12.1 SSIM 0.34	LPIPS 0.620 PSNR 13.0 SSIM 0.19	LPIPS 0.772 PSNR 15.4 SSIM 0.46	LPIPS 0.558 PSNR 15.4 SSIM 0.43
H row4 col6 10x10 brain			LPIPS 0.619 PSNR 16.5 SSIM 0.43	LPIPS 0.507 PSNR 17.2 SSIM 0.42	LPIPS 0.510 PSNR 15.1 SSIM 0.36	LPIPS 0.493 PSNR 17.8 SSIM 0.47	LPIPS 0.533 PSNR 16.7 SSIM 0.34	LPIPS 0.618 PSNR 13.4 SSIM 0.39	LPIPS 0.535 PSNR 16.8 SSIM 0.35	LPIPS 0.782 PSNR 17.8 SSIM 0.57	LPIPS 0.551 PSNR 17.8 SSIM 0.50
I row3 col3 10x10 brain			LPIPS 0.450 PSNR 21.2 SSIM 0.62	LPIPS 0.367 PSNR 17.8 SSIM 0.60	LPIPS 0.498 PSNR 19.4 SSIM 0.46	LPIPS 0.355 PSNR 20.9 SSIM 0.63	LPIPS 0.456 PSNR 21.4 SSIM 0.52	LPIPS 0.708 PSNR 14.4 SSIM 0.42	LPIPS 0.683 PSNR 19.3 SSIM 0.50	LPIPS 0.872 PSNR 10.6 SSIM 0.34	LPIPS 0.784 PSNR 17.6 SSIM 0.42
B row7 col1 10x10 heart			LPIPS 0.544 PSNR 13.7 SSIM 0.30	LPIPS 0.472 PSNR 13.9 SSIM 0.28	LPIPS 0.478 PSNR 13.0 SSIM 0.26	LPIPS 0.463 PSNR 13.9 SSIM 0.34	LPIPS 0.644 PSNR 14.1 SSIM 0.29	LPIPS 0.665 PSNR 11.4 SSIM 0.31	LPIPS 0.488 PSNR 14.2 SSIM 0.30	LPIPS 0.888 PSNR 14.2 SSIM 0.40	LPIPS 0.683 PSNR 13.5 SSIM 0.37

Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.

Biological features: Spatial split (held-out region of a piece seen in training; within-specimen interpolation) (3/3)

	Condition (UA)	C&SF reference	Ours (L-BBDM)	+ A stained refs (extra input)	+ B unstained context	+ specimen label (oracle)	L-BBDM + stock VQGAN	Ours + trainable encoder	Pixel-space BBDM	cWGAN	pix2pix
C row1 col1 5x5 heart			LPIPS 0.550 PSNR 17.5 SSIM 0.38	LPIPS 0.558 PSNR 13.7 SSIM 0.39	LPIPS 0.588 PSNR 16.5 SSIM 0.43	LPIPS 0.516 PSNR 18.0 SSIM 0.42	LPIPS 0.566 PSNR 18.1 SSIM 0.35	LPIPS 0.782 PSNR 14.5 SSIM 0.38	LPIPS 0.798 PSNR 13.1 SSIM 0.42	LPIPS 0.852 PSNR 10.2 SSIM 0.34	LPIPS 0.700 PSNR 14.4 SSIM 0.43
F row2 col2 5x5 heart			LPIPS 0.536 PSNR 18.7 SSIM 0.47	LPIPS 0.542 PSNR 18.9 SSIM 0.46	LPIPS 0.544 PSNR 16.7 SSIM 0.33	LPIPS 0.546 PSNR 20.2 SSIM 0.44	LPIPS 0.536 PSNR 18.2 SSIM 0.42	LPIPS 0.913 PSNR 12.7 SSIM 0.35	LPIPS 0.726 PSNR 19.8 SSIM 0.46	LPIPS 0.693 PSNR 22.3 SSIM 0.54	LPIPS 0.530 PSNR 17.0 SSIM 0.46
J row4 col6 10x10 heart			LPIPS 0.518 PSNR 17.8 SSIM 0.39	LPIPS 0.439 PSNR 19.8 SSIM 0.48	LPIPS 0.443 PSNR 20.4 SSIM 0.48	LPIPS 0.449 PSNR 20.6 SSIM 0.52	LPIPS 0.435 PSNR 18.3 SSIM 0.43	LPIPS 0.638 PSNR 15.6 SSIM 0.41	LPIPS 0.601 PSNR 18.0 SSIM 0.47	LPIPS 1.121 PSNR 9.1 SSIM 0.34	LPIPS 0.581 PSNR 18.6 SSIM 0.47
K row1 col2 10x10 heart			LPIPS 0.679 PSNR 14.3 SSIM 0.34	LPIPS 0.472 PSNR 16.4 SSIM 0.49	LPIPS 0.440 PSNR 17.3 SSIM 0.54	LPIPS 0.528 PSNR 16.9 SSIM 0.54	LPIPS 0.597 PSNR 16.5 SSIM 0.34	LPIPS 0.723 PSNR 12.9 SSIM 0.42	LPIPS 0.708 PSNR 13.4 SSIM 0.40	LPIPS 0.891 PSNR 9.7 SSIM 0.34	LPIPS 0.754 PSNR 13.7 SSIM 0.49
K row1 col2 5x5 heart			LPIPS 0.607 PSNR 17.7 SSIM 0.40	LPIPS 0.473 PSNR 20.1 SSIM 0.53	LPIPS 0.472 PSNR 19.0 SSIM 0.47	LPIPS 0.521 PSNR 17.7 SSIM 0.53	LPIPS 0.532 PSNR 19.0 SSIM 0.41	LPIPS 0.723 PSNR 14.9 SSIM 0.46	LPIPS 0.872 PSNR 12.1 SSIM 0.45	LPIPS 0.933 PSNR 11.6 SSIM 0.34	LPIPS 0.624 PSNR 21.0 SSIM 0.57

Tiles chosen by an author viewing only the input and reference images (no model outputs or scores); every chosen tile is shown. All diffusion panels are draw 0 (seed 1234). Per-panel metrics are for the panel shown.